A Two-Seller Formula and a Finite Reduction for Minimization Fractional Prophet Inequalities

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Abstract

Qin, Vardi, and Wierman introduced a minimization fractional prophet inequality model for sequential procurement and left open the exact competitive ratio without in-house production for general monomial exponent, number of sellers, and bounded cost support. Their paper solves the special case of two sellers and quadratic costs. This note gives two pieces of progress on that open problem. First, for two sellers, it computes the exact competitive ratio for every monomial exponent $p > 1$. If the seller cost coefficients are independently supported on $[\ell, u]$, then, with $s = \sqrt{u/\ell}$ and $\theta = 1/(p-1)$,

$$\mathrm{CR}_{p,2}(\ell, u) = \frac{s+1}{2s}\left(\frac{1+s^\theta}{2}\right)^{1/\theta}.$$

The maximizing instance has the first cost deterministic at $\sqrt{\ell u}$ and the second cost supported on the two endpoints ℓ, u with mean $\sqrt{\ell u}$. Second, for arbitrary n, the infinite-dimensional distributional supremum is reduced exactly to a compact finite-dimensional nonlinear program in which the first seller is deterministic, each interior seller has at most two atoms, and the last seller has endpoint support. Thus no continuous distributions are needed for the worst case.

1 Introduction

This note concerns the minimization fractional prophet inequality for sequential procurement studied by Qin, Vardi, and Wierman [1]. A decision maker must procure one unit of a divisible commodity from n sellers arriving sequentially. Seller i has cost coefficient C_i, drawn independently from a known distribution, and if the decision maker buys amount $x_i \geq 0$ from seller i, the incurred cost is $C_i x_i^p$. The online decision maker observes C_i before choosing x_i, but must choose irrevocably and must satisfy $\sum_i x_i = 1$. The prophet sees all C_i in advance.

The case without in-house production is the difficult case in Section 4 of [1]. Qin, Vardi, and Wierman compute the exact bounded-support competitive ratio for $p = 2$ and $n = 2$, and leave open the analytic determination of the ratio as a function of (p, n, ℓ, u). They also note that their Section 4 analysis does not directly extend beyond $p = 2$.

The main result below extends their exact two-seller formula from $p = 2$ to every $p > 1$. A secondary result gives an exact finite-dimensional reduction for arbitrary n. The latter does not solve the full closed-form problem, but it removes the infinite-dimensional search over distributions.

2 Model and notation

Fix $p > 1$ and set

$$\theta = \frac{1}{p-1} > 0.$$

For positive $c_1, \dots, c_m$, define the p-parallel aggregate

$$H_m(c_1, \dots, c_m) = \left(\sum_{j=1}^{m} c_j^{-\theta} \right)^{-1/\theta}.$$

In particular,

$$H_2(a, b) = (a^{-\theta} + b^{-\theta})^{-1/\theta} = \frac{ab}{(a^\theta + b^\theta)^{1/\theta}}.$$

When $p = 2$, $\theta = 1$, and $H_2(a, b) = ab/(a + b)$, one half of the usual harmonic mean.

For a distributional input $\mathbf{D} = (D_1, \dots, D_n)$, let $C_i \sim D_i$ be independent and supported on $[\ell, u]$, where $0 < \ell \leq u < \infty$. We evaluate the optimal online dynamic program for that known input and compare it with the prophet. Thus

$$\mathrm{CR}_{p,n}(\ell, u) = \sup_{D_1,\dots,D_n \in \mathcal{P}[\ell,u]} \frac{\mathrm{ON}_{p,n}(D_1, \dots, D_n)}{\mathbb{E} H_n(C_1, \dots, C_n)},$$

where $\mathcal{P}[\ell, u]$ denotes the Borel probability measures on $[\ell, u]$ and $\mathrm{ON}_{p,n}$ is the expected cost of the optimal online policy.

Remark 2.1. If $\ell = u$, all coefficients are deterministic and the online and offline costs agree, so $\mathrm{CR}_{p,n}(\ell, \ell) = 1$. In the rest of the note, whenever endpoint probabilities such as $(u - m)/(u - \ell)$ appear, we implicitly assume $\ell < u$.

3 The optimal online dynamic program

Lemma 3.1 (Offline and online values)**.** *For every realization $c = (c_1, \dots, c_n)$,*

$$\mathrm{OPT}(c, p) = H_n(c_1, \dots, c_n).$$

Moreover, for fixed distributions $D_1, \dots, D_n$, the optimal online expected cost is B_1, where

$$B_n = \mathbb{E}[C_n]$$

and, recursively,

$$B_i = \mathbb{E}_{C_i \sim D_i} [H_2(C_i, B_{i+1})], \qquad i = n - 1, \dots, 1.$$

After observing $C_i = c_i$ and having remaining demand y, the optimal online policy buys

$$x_i = \frac{B_{i+1}^\theta}{c_i^\theta + B_{i+1}^\theta}\, y, \qquad i < n,$$

and buys all remaining demand at stage n.

Proof. For the prophet, solve

$$\min\left\{\sum_{i=1}^{n} c_i x_i^p : \sum_{i=1}^{n} x_i = 1,\ x_i \geq 0\right\}.$$

The first-order conditions give

$$p c_i x_i^{p-1} = \lambda, \qquad i = 1, \dots, n.$$

Consequently

$$x_i = \frac{c_i^{-\theta}}{\sum_{j=1}^{n} c_j^{-\theta}},$$

and substituting gives

$$\mathrm{OPT}(c, p) = \left(\sum_{j=1}^{n} c_j^{-\theta}\right)^{-1/\theta} = H_n(c_1, \dots, c_n).$$

The nonnegativity constraints do not bind because all coefficients are positive.

For the online dynamic program, let $J_i(y)$ be the optimal expected cost from stage i onward when remaining demand is y. We prove by backward induction that

$$J_i(y) = B_i y^p.$$

At the last stage,

$$J_n(y) = \mathbb{E}[C_n y^p] = B_n y^p.$$

Suppose $J_{i+1}(y) = B_{i+1} y^p$. After seeing $C_i = c_i$, the decision maker solves

$$\min_{0 \leq x \leq y} \left\{c_i x^p + B_{i+1}(y - x)^p\right\}.$$

The first-order condition is

$$c_i x^{p-1} = B_{i+1}(y - x)^{p-1},$$

so

$$x = \frac{B_{i+1}^{\theta}}{c_i^{\theta} + B_{i+1}^{\theta}}\, y.$$

The resulting value is $H_2(c_i, B_{i+1}) y^p$. Taking expectation over C_i gives

$$J_i(y) = \mathbb{E}\left[H_2(C_i, B_{i+1})\right] y^p = B_i y^p.$$

□

4 Exact finite-dimensional reduction for arbitrary n

The next result shows that the distributional supremum does not require arbitrary probability measures. The proof is a moment reduction: for each interior seller, once one scalar moment determining the online dynamic program is fixed, the worst offline expectation is achieved by a two-point distribution.

Theorem 4.1 (Finite reduction)**.** *Assume $0 < \ell < u < \infty$. The competitive ratio*

$$\mathrm{CR}_{p,n}(\ell, u) = \sup_{D_1,\dots,D_n \in \mathcal{P}[\ell,u]} \frac{B_1(D_1,\dots,D_n)}{\mathbb{E}H_n(C_1,\dots,C_n)}$$

is unchanged if the supremum is restricted to distributions of the following form:

$$D_1 = \delta_{c_1},$$

$$D_i = (1-q_i)\delta_{a_i} + q_i\delta_{b_i}, \qquad 2 \le i \le n-1,$$

and

$$D_n = (1-q_n)\delta_\ell + q_n\delta_u,$$

where $c_1, a_i, b_i \in [\ell, u]$ and $q_i \in [0,1]$.

Equivalently, the infinite-dimensional supremum equals the compact finite-dimensional maximum

$$\mathrm{CR}_{p,n}(\ell, u) = \max \frac{B_1}{\mathrm{Off}},$$

where

$$B_n = (1-q_n)\ell + q_n u,$$

$$B_i = (1-q_i)H_2(a_i, B_{i+1}) + q_i H_2(b_i, B_{i+1}), \qquad i = n-1,\dots,2,$$

$$B_1 = H_2(c_1, B_2),$$

and

$$\mathrm{Off} = \sum_{\varepsilon_2,\dots,\varepsilon_n \in \{0,1\}} \left(\prod_{i=2}^{n} q_i^{\varepsilon_i}(1-q_i)^{1-\varepsilon_i} \right) H_n(c_1, z_{2,\varepsilon_2}, \dots, z_{n,\varepsilon_n}),$$

with

$$z_{i,0} = a_i, \qquad z_{i,1} = b_i, \qquad 2 \le i \le n-1,$$

and

$$z_{n,0} = \ell, \qquad z_{n,1} = u.$$

Proof. Fix all distributions except D_k.

First consider $2 \le k \le n-1$. Let

$$\psi_k(c) = H_2(c, B_{k+1}).$$

The online numerator depends on D_k only through

$$B_k = \mathbb{E}[\psi_k(C_k)].$$

Indeed, once B_k is fixed, all earlier values $B_{k-1},\dots,B_1$ are fixed by the recursion in Theorem 3.1. The offline denominator has the form

$$\mathbb{E}[\phi_k(C_k)],$$

where

$$\phi_k(c) = \mathbb{E}\left[H_n(C_1,\dots,C_{k-1}, c, C_{k+1},\dots,C_n)\right],$$

and the expectation is over all variables except C_k.

Since $c \mapsto \psi_k(c) = H_2(c, B_{k+1})$ is continuous and strictly increasing on $[\ell, u]$, it is a homeomorphism from $[\ell, u]$ onto its image. Write $t = \psi_k(c)$ and

$$\eta_k(t) = \phi_k(\psi_k^{-1}(t)).$$

Preserving B_k is the same as preserving the mean of t. Among probability distributions on the compact interval $\psi_k([\ell, u])$ with fixed mean B_k, the minimum possible value of $\mathbb{E}[\eta_k(t)]$ is attained on the lower convex envelope of η_k. A point of this lower convex envelope is either a point of the graph of η_k or lies on a chord between two points of that graph. Therefore the minimizing distribution has support of size at most two. Pulling back by ψ_k^{-1}, D_k can be replaced by a distribution supported on at most two points, preserving the online numerator and weakly decreasing the offline denominator.

Next consider $k = n$. The online numerator depends on D_n only through

$$B_n = \mathbb{E}[C_n].$$

For fixed $c_1, \ldots, c_{n-1}$, the map

$$c \mapsto H_n(c_1, \ldots, c_{n-1}, c) = (A + c^{-\theta})^{-1/\theta}, \qquad A = \sum_{i=1}^{n-1} c_i^{-\theta},$$

is concave on $(0, \infty)$, because

$$\frac{d^2}{dc^2}(A + c^{-\theta})^{-1/\theta} = -(\theta + 1)Ac^{-\theta-2}(A + c^{-\theta})^{-1/\theta-2} < 0.$$

Thus, among distributions on $[\ell, u]$ with fixed mean B_n, the expectation of this concave function is minimized by the endpoint distribution

$$C_n = \begin{cases} \ell, & \text{with probability } (u - B_n)/(u - \ell), \\ u, & \text{with probability } (B_n - \ell)/(u - \ell). \end{cases}$$

This preserves the online numerator and weakly decreases the offline denominator.

Finally consider $k = 1$. With $D_2, \ldots, D_n$ fixed, the ratio is

$$\frac{\mathbb{E}[\psi_1(C_1)]}{\mathbb{E}[\phi_1(C_1)]},$$

where

$$\psi_1(c) = H_2(c, B_2), \qquad \phi_1(c) = \mathbb{E}[H_n(c, C_2, \ldots, C_n)].$$

Since $\phi_1(c) > 0$, this ratio is a weighted average of the pointwise ratios $\psi_1(c)/\phi_1(c)$ with weights proportional to $\phi_1(c)\, \mathrm{d}D_1(c)$. Hence it is at most

$$\max_{c \in [\ell, u]} \frac{\psi_1(c)}{\phi_1(c)}.$$

Thus D_1 may be replaced by a point mass.

Applying these replacements successively, from D_n down to D_2, and then to D_1, never decreases the ratio. This proves that the supremum is unchanged under the stated finite-support restriction. The displayed finite-dimensional optimization is the ratio written out under that restriction. Its parameter set is compact and its denominator is positive, so the supremum in the reduced problem is a maximum. $\square$

Remark 4.2. The theorem does not assert that all two-point distributions must use the endpoints. It proves that arbitrary distributions are unnecessary. The remaining all-n problem is the nonlinear finite-dimensional maximization displayed above.

5 The exact two-seller ratio for every $p > 1$

Theorem 5.1 (Two-seller formula)**.** *Let $p > 1$, $0 < \ell \le u < \infty$, and set*

$$\theta = \frac{1}{p-1}, \qquad s = \sqrt{\frac{u}{\ell}}.$$

Then, in the two-seller problem without in-house production,

$$\boxed{\mathrm{CR}_{p,2}(\ell, u) = \frac{s+1}{2s}\left(\frac{1+s^{\theta}}{2}\right)^{1/\theta}}.$$

Equivalently,

$$\boxed{\mathrm{CR}_{p,2}(\ell, u) = \frac{1+\sqrt{\ell/u}}{2}\left(\frac{1+(u/\ell)^{1/(2(p-1))}}{2}\right)^{p-1}}.$$

For $\ell < u$, the maximizing distributions are

$$D_1 = \delta_{\sqrt{\ell u}}$$

and

$$D_2 = \frac{\sqrt{u}}{\sqrt{u}+\sqrt{\ell}}\,\delta_\ell + \frac{\sqrt{\ell}}{\sqrt{u}+\sqrt{\ell}}\,\delta_u.$$

When $\ell = u$, the ratio is 1 and the instance is deterministic.

Proof. The ratio is invariant under multiplying every cost coefficient by the same positive constant. We therefore normalize $\ell = 1$ and write $u = v > 1$. Set

$$h(a, b) = H_2(a, b) = (a^{-\theta} + b^{-\theta})^{-1/\theta}.$$

By Theorem 4.1, for $n = 2$ it suffices to take

$$D_1 = \delta_c$$

and

$$D_2 = \frac{v-m}{v-1}\delta_1 + \frac{m-1}{v-1}\delta_v,$$

where $c, m \in [1, v]$. Here $m = \mathbb{E}[C_2]$. The competitive ratio of this instance is

$$R(c, m) = \frac{h(c, m)}{\frac{v-m}{v-1}h(c, 1) + \frac{m-1}{v-1}h(c, v)}.$$

We prove

$$\max_{1 \le c, m \le v} R(c, m) = R(\sqrt{v}, \sqrt{v}).$$

Fix c. Define

$$A = h(c, 1), \qquad B = h(c, v), \qquad \rho = \frac{B}{A},$$

and

$$d = \frac{v-\rho}{\rho-1}.$$

Since $h(c,\cdot)$ is strictly increasing and $1 < v$, we have $1 < \rho < v$ and $d > 0$. The denominator of $R(c,m)$ is

$$\begin{aligned}\frac{v-m}{v-1}A + \frac{m-1}{v-1}B &= \frac{A}{v-1}\big((v-m) + (m-1)\rho\big)\\ &= \frac{A(\rho-1)}{v-1}(m+d).\end{aligned}$$

Therefore

$$\frac{\partial}{\partial m}\log R(c,m) = \frac{c^\theta}{m(c^\theta+m^\theta)} - \frac{1}{m+d} = \frac{c^\theta d - m^{\theta+1}}{m(c^\theta+m^\theta)(m+d)}.$$

For fixed c, the only possible interior maximizer is therefore

$$m_c = (c^\theta d)^{1/(\theta+1)}.$$

If this point lies outside $[1,v]$, then the maximum over $m \in [1,v]$ occurs at an endpoint and equals 1, since $R(c,1) = R(c,v) = 1$. Such a value cannot be globally optimal because $R(\sqrt{v},\sqrt{v}) > 1$; equivalently, the power mean $((1+s^\theta)/2)^{1/\theta}$ of 1 and $s = \sqrt{v}$ strictly exceeds their harmonic mean $2s/(s+1)$. Thus any global maximizer with value larger than 1 must occur at $m = m_c$.

It remains to maximize

$$F(c) = R(c, m_c)$$

over the values of c for which $m_c \in [1,v]$. We compute the sign of $F'(c)$. First note that

$$\rho = v\left(\frac{c^\theta+1}{c^\theta+v^\theta}\right)^{1/\theta}.$$

Hence ρ is strictly increasing in c, and

$$\rho(\sqrt{v}) = \sqrt{v}.$$

Put $a = c^\theta$. From the formula for ρ,

$$\rho^\theta = v^\theta\frac{a+1}{a+v^\theta}.$$

Solving for a gives

$$a = v^\theta\frac{\rho^\theta-1}{v^\theta-\rho^\theta}, \qquad a+1 = \rho^\theta\frac{v^\theta-1}{v^\theta-\rho^\theta},$$

and the defining relation for m_c gives

$$m_c^{\theta+1} = ad = v^\theta\frac{\rho^\theta-1}{v^\theta-\rho^\theta}\frac{v-\rho}{\rho-1}.$$

By the envelope theorem,

$$\frac{d}{dc}\log F(c) = \frac{\partial}{\partial c}\log R(c,m)\bigg|_{m=m_c}.$$

A direct algebraic simplification, recorded in Theorem A.1, gives

$$c\frac{d}{dc}\log F(c) = \frac{(\rho^{\theta+1}-1)(N(\rho)-m_c)}{(a+1)(\rho-1)(m_c+d)},$$

where

$$N(\rho) = \frac{\rho^\theta - 1}{\rho^{\theta+1} - 1} \frac{v^{\theta+1} - \rho^{\theta+1}}{v^\theta - \rho^\theta}.$$

All factors outside $N(\rho) - m_c$ are positive. Thus the sign of $F'(c)$ is the sign of $N(\rho) - m_c$.

We compare $N(\rho)$ and m_c by comparing their $(\theta + 1)$-st powers. Define

$$\Delta(\rho) = \log \frac{N(\rho)^{\theta+1}}{m_c^{\theta+1}}.$$

Substitution of the displayed formulas for $N(\rho)$ and $m_c^{\theta+1}$ yields

$$\begin{aligned}\Delta(\rho) = \theta \log(\rho^\theta - 1) &- (\theta + 1) \log(\rho^{\theta+1} - 1) \\ &+ (\theta + 1) \log(v^{\theta+1} - \rho^{\theta+1}) - \theta \log(v^\theta - \rho^\theta) \\ &- \theta \log v - \log(v - \rho) + \log(\rho - 1).\end{aligned}$$

Now write

$$\rho = e^{2x}, \qquad v/\rho = e^{2y}.$$

Equivalently,

$$x = \frac{1}{2} \log \rho, \qquad y = \frac{1}{2} \log(v/\rho).$$

After cancelling exponential factors,

$$\Delta(\rho) = \Gamma_\theta(x) - \Gamma_\theta(y),$$

where

$$\Gamma_\theta(t) = \log \sinh t + \theta \log \sinh(\theta t) - (\theta + 1) \log \sinh((\theta + 1)t).$$

We claim that Γ_θ is strictly decreasing on $(0, \infty)$. Indeed,

$$\Gamma'_\theta(t) = \coth t + \theta^2 \coth(\theta t) - (\theta + 1)^2 \coth((\theta + 1)t).$$

For fixed $t > 0$, the function $q \mapsto q \coth(qt)$ is strictly increasing on $(0, \infty)$, because

$$\frac{d}{dq}\big(q \coth(qt)\big) = \coth(qt) - qt \operatorname{csch}^2(qt) = \frac{\sinh(qt) \cosh(qt) - qt}{\sinh^2(qt)} > 0.$$

Therefore

$$(\theta + 1)\big((\theta + 1) \coth((\theta + 1)t)\big) > \coth t + \theta\big(\theta \coth(\theta t)\big),$$

which is exactly $\Gamma'_\theta(t) < 0$.

It follows that

$$\Delta(\rho) > 0 \iff x < y \iff \rho < \sqrt{v},$$

with equality precisely when $\rho = \sqrt{v}$, and with the opposite inequality for $\rho > \sqrt{v}$. Since $N(\rho)$ and m_c are positive, the sign of $N(\rho) - m_c$ is the sign of $\Delta(\rho)$. Because $\rho(c)$ is strictly increasing and $\rho(\sqrt{v}) = \sqrt{v}$, we conclude that

$$F'(c) > 0 \quad \text{for } c < \sqrt{v}, \qquad F'(c) < 0 \quad \text{for } c > \sqrt{v}.$$

Thus the global maximum is attained at $c = \sqrt{v}$. The corresponding maximizing mean is also $m = \sqrt{v}$.

Let $s = \sqrt{v}$. Then

$$h(s,s) = s\,2^{-1/\theta},$$

and

$$\frac{v-s}{v-1} = \frac{s}{s+1}, \qquad \frac{s-1}{v-1} = \frac{1}{s+1}.$$

Moreover,

$$h(s,1) = \frac{s}{(1+s^\theta)^{1/\theta}}, \qquad h(s,v) = \frac{s^2}{(1+s^\theta)^{1/\theta}}.$$

Therefore the denominator at $c = m = s$ is

$$\frac{s}{s+1}h(s,1) + \frac{1}{s+1}h(s,v) = \frac{2s^2}{(s+1)(1+s^\theta)^{1/\theta}}.$$

Consequently

$$\begin{aligned} R(s,s) &= \frac{s\,2^{-1/\theta}}{2s^2/((s+1)(1+s^\theta)^{1/\theta})} \\ &= \frac{s+1}{2s}\left(\frac{1+s^\theta}{2}\right)^{1/\theta}. \end{aligned}$$

This proves the normalized formula.

Returning to general $[\ell, u]$, scaling all costs by ℓ multiplies both online and offline costs by ℓ, hence does not change the ratio. Since $v = u/\ell$ and $s = \sqrt{u/\ell}$, the formula becomes

$$\mathrm{CR}_{p,2}(\ell,u) = \frac{s+1}{2s}\left(\frac{1+s^\theta}{2}\right)^{1/\theta}.$$

Using $1/\theta = p-1$ gives the equivalent expression in the theorem statement.

The maximizing normalized instance is

$$C_1 = s,$$

and

$$C_2 = \begin{cases} 1, & \text{with probability } s/(s+1), \\ s^2, & \text{with probability } 1/(s+1). \end{cases}$$

Rescaling by ℓ gives

$$D_1 = \delta_{\sqrt{\ell u}},$$

and

$$D_2 = \frac{\sqrt{u}}{\sqrt{u}+\sqrt{\ell}}\,\delta_\ell + \frac{\sqrt{\ell}}{\sqrt{u}+\sqrt{\ell}}\,\delta_u.$$

□

Corollary 5.2 (Recovery of the quadratic two-seller formula)**.** *For $p = 2$, $\theta = 1$, and Theorem 5.1 gives*

$$\mathrm{CR}_{2,2}(\ell,u) = \frac{s+1}{2s}\frac{1+s}{2} = \frac{1}{4}\left(s + \frac{1}{s}\right) + \frac{1}{2}, \qquad s = \sqrt{\frac{u}{\ell}}.$$

Equivalently,

$$\mathrm{CR}_{2,2}(\ell,u) = \frac{1}{4}\left(\sqrt{\frac{u}{\ell}} + \sqrt{\frac{\ell}{u}}\right) + \frac{1}{2},$$

which is Theorem 3 of Qin, Vardi, and Wierman [1].

6 Concluding remarks

The formula in Theorem 5.1 resolves the two-seller bounded-support problem for every exponent $p > 1$. It shows that the worst instance retains the same qualitative form as in the quadratic case: a deterministic first seller at the geometric mean of the support endpoints, followed by an endpoint-supported second seller whose mean is the same geometric mean.

For arbitrary n, Theorem 4.1 gives an exact compact nonlinear program. This is a genuine finite reduction, but it is not a closed form for the competitive ratio. A natural next step is to determine when the two atoms in the interior distributions of Theorem 4.1 can be forced to be endpoints, or when they collapse to point masses. Such a result would turn the remaining open problem into a much smaller combinatorial-continuous optimization over endpoint "long-shot" distributions and deterministic sellers.

A Algebra for the derivative identity

Lemma A.1. *In the notation of the proof of Theorem 5.1, for*

$$F(c) = R(c, m_c), \qquad m_c^{\theta+1} = a\,\frac{v-\rho}{\rho-1}, \qquad a = c^\theta,$$

one has

$$c\frac{d}{dc}\log F(c) = \frac{(\rho^{\theta+1}-1)(N(\rho)-m_c)}{(a+1)(\rho-1)(m_c+d)},$$

where

$$d = \frac{v-\rho}{\rho-1}$$

and

$$N(\rho) = \frac{\rho^\theta - 1}{\rho^{\theta+1}-1}\,\frac{v^{\theta+1}-\rho^{\theta+1}}{v^\theta - \rho^\theta}.$$

Proof. Recall

$$R(c,m) = \frac{h(c,m)}{A(\rho-1)(m+d)/(v-1)},$$

where

$$h(c,m) = \frac{cm}{(c^\theta + m^\theta)^{1/\theta}}, \qquad A = h(c,1), \qquad \rho = \frac{h(c,v)}{h(c,1)}.$$

By the envelope theorem,

$$c\frac{d}{dc}\log F(c) = c\,\frac{\partial}{\partial c}\log R(c,m)\bigg|_{m=m_c}.$$

At fixed m,

$$c\frac{\partial}{\partial c}\log h(c,m) = \frac{m^\theta}{a+m^\theta},$$

and

$$c\frac{d}{dc}\log A = \frac{1}{a+1}.$$

Also

$$\rho = v\left(\frac{a+1}{a+v^\theta}\right)^{1/\theta},$$

so

$$c\frac{d\rho}{dc} = \rho\,\frac{a(v^\theta - 1)}{(a+1)(a+v^\theta)}.$$

Using

$$d = \frac{v-\rho}{\rho-1}, \qquad \frac{dd}{d\rho} = -\frac{v-1}{(\rho-1)^2},$$

we obtain

$$c\frac{d}{dc}\log F(c) = \frac{m_c^\theta}{a+m_c^\theta} - \frac{1}{a+1} - \frac{c\rho'}{\rho-1} - \frac{cd'}{m_c+d}.$$

Here and below primes denote derivatives with respect to c, and $d' = (dd/d\rho)\rho'$. At the optimizing value m_c, the first-order condition in m gives

$$m_c^{\theta+1} = ad.$$

This implies

$$\frac{m_c^\theta}{a+m_c^\theta} = \frac{d}{m_c+d}.$$

Furthermore, substituting the expression for $c\rho'$ and using

$$a = v^\theta\,\frac{\rho^\theta - 1}{v^\theta - \rho^\theta}, \qquad a+1 = \rho^\theta\,\frac{v^\theta - 1}{v^\theta - \rho^\theta},$$

we get the useful identity

$$c\rho' = \rho\frac{a(v^\theta - 1)}{(a+1)(a+v^\theta)} = \frac{\rho^{1-\theta}(\rho^\theta - 1)(v^\theta - \rho^\theta)}{v^\theta - 1}.$$

Combining these identities and simplifying to a common denominator gives

$$c\frac{d}{dc}\log F(c) = \frac{1}{(a+1)(\rho-1)(m_c+d)}\left[(\rho^\theta - 1)\frac{v^{\theta+1} - \rho^{\theta+1}}{v^\theta - \rho^\theta} - m_c(\rho^{\theta+1} - 1)\right].$$

Factoring out $\rho^{\theta+1} - 1$ from the bracket gives exactly

$$c\frac{d}{dc}\log F(c) = \frac{(\rho^{\theta+1} - 1)(N(\rho) - m_c)}{(a+1)(\rho-1)(m_c+d)}.$$

□

References

[1] Junjie Qin, Shai Vardi, and Adam Wierman. Minimization fractional prophet inequalities for sequential procurement. *Mathematics of Operations Research*, 49(2):928–947, 2024. doi:10.1287/moor.2021.0173.